

Assignment – 17
(Multi-Level Inheritance)

1. Create a class *Number* :

Data member: An array of type integer.

Constructor: Constructor with one parameter n, that is the size of the array.

Allocate n memory for the array and input n numbers into the array.

Method-1: To display all the values in the array.

Derive a class *OddNum* from the class *Number*:

Data member: An array of type integer.

Constructor: To count the odd numbers present in the array of its base class *Number* and accordingly allocate memory for its own array.

Method-1: To copy all odd numbers from its base class array to its own array.

Method-2: To display all odd numbers.

Derive a class *PrimeNum* from the class *OddNum*:

Data member: An array of type integer.

Constructor: To count the prime numbers present in the array of its base class *OddNum* and accordingly allocate memory for its own array.

Method-1: To copy all prime numbers from its base class array to its own array.

Method-2: To display all prime numbers.

2. Define a class ***Employee*** having private members – id, name, department, salary. Define default and parameterized constructors. Create a subclass called “***Manager***” with private member bonus. Define methods ***accept()*** and ***display()*** in both the classes. Create *n* objects of the ***Manager*** class and display the details of the manager having the maximum total salary (salary+bonus).